

MODULE 3: PROGRAMMING FOR BIOLOGICAL DATA

Session 7: Diagnostic Accuracy

Sensitivity, Specificity, and ROC Curves

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Learning Objectives



Confusion Matrix

Understand components: TP, FP, TN, FN.



Metrics

Calculate Sensitivity, Specificity, PPV, and NPV.



ROC & AUC

Generate and interpret ROC curves and AUC values.



Optimization

Find optimal cutoffs using the pROC package.

Why Diagnostic Accuracy Matters

The Core Question

"Is this patient positive or negative?"

No test is perfect. We must quantify:

- **Missed cases (False Negatives):** Can lead to delayed treatment and spread of disease.
- **False alarms (False Positives):** Can lead to unnecessary anxiety and overtreatment.

Risk of FN

Patient sent home while sick.

Risk of FP

Healthy patient treated needlessly.

Part 1: The Confusion Matrix

Foundations of Diagnostic Testing

The Four Possible Outcomes

	Actually Positive	Actually Negative
Test Positive	✓ True Positive (TP)	✗ False Positive (FP)
Test Negative	✗ False Negative (FN)	✓ True Negative (TN)

	Predicted: NO	Predicted: YES	
n = 200			
Actual: NO	TN = 35	FP = 10	45
Actual: YES	FN = 5	TP = 150	155
	40	160	200

Example: Dengue Rapid Test

Tested 100 patients against gold standard PCR.

	PCR+ (Disease)	PCR- (Healthy)	Total
Rapid Test +	45 (TP)	5 (FP)	50
Rapid Test -	8 (FN)	42 (TN)	50
Total	53	47	100

Sensitivity (True Positive Rate)

Definition

"How good is the test at finding POSITIVES?"

Also known as **Recall**.

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Calculation

Using our Dengue Example:

$$\frac{45}{45 + 8} = 84.9\%$$

The test detects ~85% of true cases.

Clinical Meaning: SnNout



Rule OUT

If a highly sensitive test is **NEGATIVE**, you can be confident the patient does NOT have the disease.

SnNout

Sensitivity

Negative

out

Example: Dengue Rapid Test

Tested 100 patients against gold standard PCR.

	PCR+ (Disease)	PCR- (Healthy)	Total
Rapid Test +	45 (TP)	5 (FP)	50
Rapid Test -	8 (FN)	42 (TN)	50
Total	53	47	100

Specificity (True Negative Rate)

Definition

"How good is the test at identifying NEGATIVES?"

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

Calculation

Using our Dengue Example:

$$\frac{42}{42 + 5} = 89.4\%$$

The test correctly identifies ~89% of healthy people.

Clinical Meaning: SpPin



Rule IN

If a highly specific test is **POSITIVE**, you can be confident the patient **HAS** the disease.

False Positive low, then should be True positive.

SpPin

Specificity

Positive

in

Example: Dengue Rapid Test

Tested 100 patients against gold standard PCR.

	PCR+ (Disease)	PCR- (Healthy)	Total
Rapid Test +	45 (TP)	5 (FP)	50
Rapid Test -	8 (FN)	42 (TN)	50
Total	53	47	100

Predictive Values

Crucial Note: These values depend heavily on disease *prevalence*.

PPV (Positive Predictive Value)

If test is +, how likely is disease?

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Example: 90%

NPV (Negative Predictive Value)

If test is -, how likely is healthy?

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}}$$

Example: 84%

The Effect of Prevalence

Same test (Sens 90%, Spec 90%), different populations:

Scenario	Prevalence	PPV (Confidence in +)	NPV (Confidence in -)
Screening	1%	8.3% (Very Low!)	99.9% (High)
Hospital	50%	90.0% (Moderate)	90.0% (Moderate)
Epidemic	90%	98.8% (High)	52.9% (Low)

Calculating Metrics in R

```
TP ← 45; FP ← 5  
FN ← 8;  TN ← 42  
  
sens ← TP / (TP + FN)  
spec ← TN / (TN + FP)  
ppv  ← TP / (TP + FP)  
npv  ← TN / (TN + FN)  
  
cat("Sens:", sens)  
cat("Spec:", spec)
```

Manual Verification

It is often easiest to define the variables directly from your confusion matrix (table) output and run the formulas.

This ensures you don't mix up the denominators.

Sensitivity-Specificity Trade-off

You can't maximize both!

Diagnostic tests usually output a continuous value (e.g., OD, concentration).

Moving the cutoff:

- **Lower Cutoff:** Catch more disease ($\uparrow$ Sens) but more false alarms ($\downarrow$ Spec).
- **Higher Cutoff:** Miss more disease ($\downarrow$ Sens) but fewer false alarms ($\uparrow$ Spec).



Clinical Priorities

Screening



Priority: High Sensitivity

Don't miss any cases. Accept some false positives.

Confirmation



Priority: High Specificity

Avoid false diagnosis. Be sure before treating.

Blood Bank



Priority: Both High

Safety critical. Cannot miss HIV/HepC, cannot waste blood.

Part 2: ROC Analysis

Receiver Operating Characteristic

What is an ROC Curve?

A graph that visualizes the performance of a test across **all possible cutoffs**.

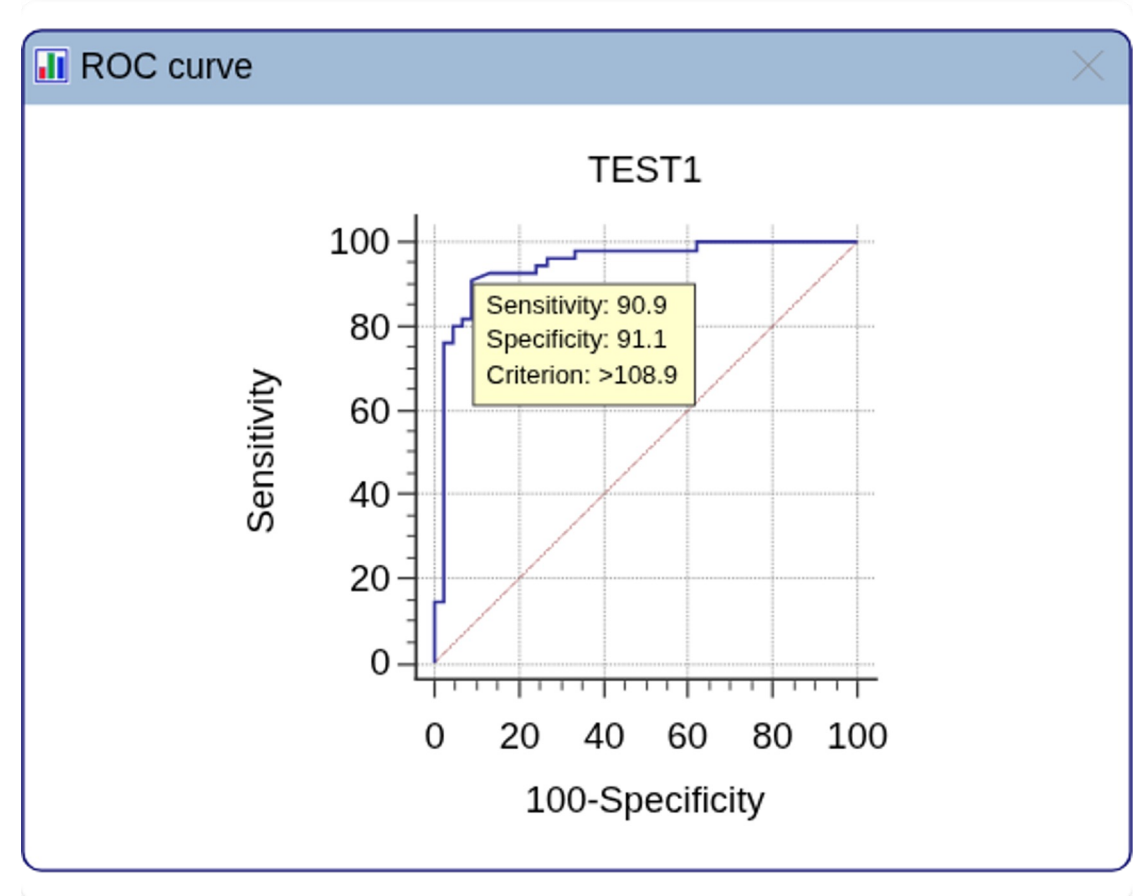
The Axes

- **Y-Axis:** Sensitivity (True Positive Rate)
- **X-Axis:** $1 - \text{Specificity}$ (False Positive Rate)

Reading an ROC Curve

Key Landmarks

- **Upper Left Corner:** The "Perfect" Test (100% Sens, 100% Spec).
- **Diagonal Line:** Random Guessing (no better than a coin flip).
- **The Curve:** The closer it bows toward the upper left, the better the test.



AUC: Area Under the Curve

Definition

A single number that summarizes the overall discrimination ability of the test.

Probability that a random positive has a higher score than a random negative.

Interpretation

AUC	Meaning
0.5	No discrimination (Random)
0.7-0.8	Acceptable
0.8-0.9	Excellent
>0.9	Outstanding

Using the pROC Package

```
library(pROC)

# Generate ROC object
roc_obj ← roc(response = actual_status,
               predictor = test_values)

# Calculate AUC
auc(roc_obj)

# Plot
plot(roc_obj)
```

Workflow

1. Load pROC.
2. Create ROC object using truth vector and test result vector.
3. Extract AUC.
4. Visualize.

Finding the Optimal Cutoff

Youden Index

Method to find the "best balance" between sensitivity and specificity.

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

Maximizes distance from the diagonal line.

R Code

```
# Get best coords
coords(roc_obj, "best",
      ret = c("threshold",
              "sensitivity",
              "specificity"))
```

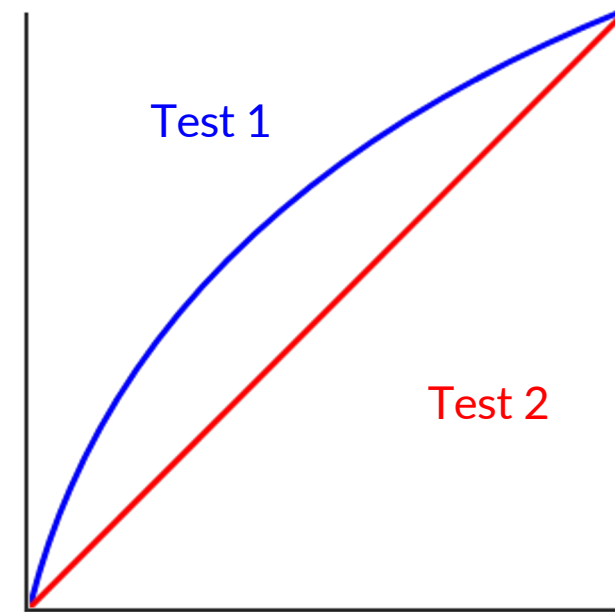
Comparing Two Tests

```
roc1 <- roc(actual, test1)
roc2 <- roc(actual, test2)

plot(roc1, col="blue")
plot(roc2, col="red", add=TRUE)

# Compare AUCs
auc(roc1)
auc(roc2)
```

When curves overlap, AUC helps decide the overall winner. If curves cross, one test might be better for high sensitivity while the other is better for high specificity.



Key Takeaways

- **Confusion Matrix:** The foundation (TP, FP, FN, TN).
- **Sensitivity:** Finding positives (Rule OUT).
- **Specificity:** Finding negatives (Rule IN).

- **ROC Curve:** Visualizes performance across all cutoffs.
- **AUC:** The grade of the test (0.5 to 1.0).
- **Optimal Cutoff:** Depends on clinical context (Screening vs Confirmation).

Tutorial 7

Optimizing the Cutoff

Open your R environment

Tutorial 7: Optimizing the Cutoff

ROC Analysis in Practice

Your Scenario

Lab Context

Your lab is evaluating a **new dengue ELISA assay**.

- **Data:** 40 patient samples.
- **Measurements:** ELISA OD values (higher = more positive).
- **Gold Standard:** PCR results (0/1).

Goal: Find the optimal OD cutoff for reporting.



Tutorial Steps

1

Load

Load the data
frame.

2

ROC

Generate ROC
object.

3

Plot

Visualize curve &
AUC.

4

Cutoff

Find Youden Index.

Step 1: Load Data

```
# Create dataset
elisa <- data.frame(
  OD = c(0.85, 0.32, 1.45, ... ),
  PCR = c(1, 0, 1, ... )
)

# Preview
head(elisa)
```

Ensure your data has a predictor column (OD) and a binary response column (PCR).

Step 2: Generate ROC

Code

```
library(pROC)

roc_res ← roc(elisa$PCR, elisa$OD)

# View AUC
auc(roc_res)
```

Note

The roc() function needs:

1. **Response:** The truth (PCR).
2. **Predictor:** The test value (OD).

Step 3: Plot ROC

```
plot(roc_res,  
     print.auc = TRUE,  
     col = "steelblue",  
     lwd = 3,  
     main = "Dengue ELISA ROC")
```

Checklist

- Is the curve above the diagonal?
- Is the AUC > 0.8?
- Does it look smooth or jagged? (Small N = jagged)

Step 4: Find Optimal Cutoff

Code

```
coords(roc_res, "best",  
      ret = c("threshold",  
              "sensitivity",  
              "specificity"))
```

Output Interpretation

R will output the threshold value that maximizes Sensitivity + Specificity.

Example Output:

Threshold: 0.65

Sens: 0.95

Spec: 0.90

Step 5: Final Recommendation

Based on your analysis, you report to the lab director.

"We recommend an OD cutoff of 0.65. This provides a sensitivity of 95% and specificity of 90%, optimizing our ability to detect Dengue while minimizing false positives."

Next Steps

Proceed to Assessment 2 to test your knowledge!

Session 7 Complete

You are now ready for Assessment 2.